Ghazaleh Asghari PhD

Research Mathematician · Seeking Quantitative Research Roles

Reading, UK · ghazalasghari@gmail.com

UK Global Talent visa (Exceptional Promise): no employer sponsorship required

PROFILE

Research mathematician (PhD, University of Reading, 2026) moving into quantitative research. Specialises in operator theory and complex analysis, with two published papers and one submitted preprint. Research experience includes spectral estimates, Gaussian-type weighted function spaces and semiclassical asymptotics. PhD-level coursework in probability and stochastic processes; Python for algorithmic problem solving. Professional experience reviewing advanced mathematical arguments for correctness and unstated assumptions.

QUANTITATIVE AND COMPUTATIONAL SKILLS

Probability & stochastic processes	PhD-level coursework: Markov chains, Poisson processes, Brownian motion, martingales, stopping times, Itô calculus. Extensive independent problem solving in probability, conditional expectation and combinatorics.
Analysis & asymptotics	Spectral theory of operators; singular values and Schatten norms; stationary-phase and Laplace-type asymptotics; Gaussian kernels and reproducing-kernel Hilbert spaces.
Programming	Python for algorithmic problem solving: data structures, complexity analysis, translating mathematical arguments into code. Working familiarity with NumPy, pandas, Matplotlib and Jupyter; exposure to C++.
Data & tools	Regression, curve fitting and quantitative analysis of experimental data (physics laboratory work). Extensive LaTeX.

PROFESSIONAL EXPERIENCE

Scientific Reviewer, AI-Generated Mathematics and Physics Mar 2026 – present
PI1 Labs, remote

- Review research-level mathematics and theoretical physics produced by AI systems for correctness, assumptions, logical consistency and rigour.
- Identify errors and gaps in technical arguments and independently judge whether conclusions are mathematically justified.

Doctoral Researcher, Department of Mathematics and Statistics 2022 – 2026
University of Reading, UK

- Co-authored two published papers and one submitted preprint on Toeplitz and Hankel operators and Berezin transforms.
- Presented at Oxford, King's College London, Lille, Toledo (USA) and IWOTA 2025; referee for three journals.

EDUCATION

PhD, Mathematics 2022 – 2026
University of Reading, UK. Awarded May 2026. Advisor: Prof. Jani Virtanen.
Thesis: *Toeplitz, Hankel, and Berezin transforms on Fock-type spaces*. Spectral properties of projection-based operators and Gaussian-type averaging transforms on weighted reproducing-kernel spaces of holomorphic functions.

MSc, Mathematics 2018 – 2021
University of Zurich (UZH) and ETH Zurich, Switzerland. Advisor: Prof. Will J. Merry.
Thesis: *Fidelity and transversely intersecting Lagrangians*. Derived leading-order asymptotics for quantum-state fidelity using Bergman kernels and stationary phase; dominant contributions determined by critical points and their Hessians.

MSc, Physics 2015 – 2018
Sharif University of Technology, Tehran, Iran. Advisor: Prof. Mohammad Mehdi Sheikh-Jabbari.
Thesis: *Symplectic Symmetries and Their Associated Hilbert Spaces*. Symplectic geometry, geometric quantization and representations of infinite-dimensional symmetry groups (Virasoro, BMS_3).

BSc, Physics, Minor in Mathematics 2010 – 2015
Sharif University of Technology, Tehran, Iran. Thesis: *Causal Sets Approach to Quantum Gravity*.

PUBLICATIONS

Fixed points of the Berezin transform on Fock-type spaces 2026
with Ž. Čučković and S. Şahutoğlu. *Proc. Amer. Math. Soc.* **154** (2026), 3011–3026. arXiv:2508.13115
For weights $e^{-|z|^m}$, polynomial fixed points are harmonic outside a possible countable exceptional set of $m > 0$.

Toeplitz operators on large vector-valued Fock spaces 2025
with H. Arroussi and J. A. Virtanen. Submitted. arXiv:2504.15239
Boundedness, compactness and Schatten-class criteria for positive operator-valued symbols on Hilbert-space-valued Fock spaces.

Schatten class Hankel operators on doubling Fock spaces and the Berger–Coburn phenomenon 2024
with Z. Hu and J. A. Virtanen. *J. Math. Anal. Appl.* **540** (2024), 128596. doi arXiv:2312.06656
Characterised Schatten membership; proved the Hilbert–Schmidt Berger–Coburn property for bounded symbols and counterexamples in certain doubling Fock spaces for $0 < p \leq 1$.

In preparation: Toeplitz operators on large Banach vector-valued Fock spaces (sole author; final stages) · *Sarason's conjecture on doubling Fock spaces* (with S. Miihkinen and J. A. Virtanen).

HONOURS AND AWARDS

EPSRC PhD Scholarship 2022 – 2026
Four-year award; sole EPSRC-funded PhD student in the department's 2022 intake.

SwissMAP Equal Opportunities Excellence Scholarship 2018 – 2020

Excellence Master Fellowship, University of Geneva 2018
Awarded annually to about 16 Master's students across all disciplines.

SELECTED TALKS

Functional Analysis Seminar, Université de Lille, France — Apr 2026
Functional Analysis Seminar, University of Oxford, UK — Nov 2025
IWOTA 2025, University of Twente, Netherlands — Jul 2025
Analysis Seminar, King's College London, UK — Jun 2024
Complex Analysis & Operator Theory Seminar, University of Toledo, USA — Apr 2024

TEACHING AND ACADEMIC SERVICE

- Referee for *J. Math. Anal. Appl.*, *Complex Var. Elliptic Equ.* and *Bull. Sci. Math.* — 2023 – present
- Co-organiser, SIAM–IMA Reading Student Chapter seminar series — 2024 – 2026
- Substitute teacher, Calculus in One Variable, KTH Royal Institute of Technology, Sweden — 2023
- Teaching assistant, University of Reading: analysis, analytic number theory, foundations — 2022 – 2024
- Designed and taught a semester seminar on *Topology from the Differentiable Viewpoint* — 2022